AUTOMOBILE CAR DATASET

FULL EXPLORATORY DATA ANALYSIS PROJECT

Professional project report generated from the uploaded **project1.ipynb** notebook and Automobile_data.csv dataset.

Project	Automobile Car Dataset — EDA
Dataset	Automobile_data.csv
Notebook	project1.ipynb
Dataset records	205
Original dataset columns	15
Target variable	price
Analysis	Univariate • Bivariate • Multivariate
Tools	Python • Pandas • NumPy • Matplotlib • Seaborn • Scikit-learn

Prepared as a complete project document containing methodology, cleaning, preprocessing, all notebook visualizations, analytical tables, key insights, and conclusion.

1. Project Overview

The objective of this project is to perform Exploratory Data Analysis (EDA) on an automobile dataset. The notebook follows a structured workflow covering dataset import, missing-value inspection, cleaning, numerical and categorical data preparation, skewness analysis, transformation, scaling, feature engineering, visualization, relationship analysis, and final insights.

The target variable selected in the notebook is **price**. The final feature-target split is represented by X and y, respectively.

2. Dataset & Data Quality

The supplied notebook loads Automobile_data.csv and contains 205 records. The original dataset has 15 columns. The notebook first checks missing values and then checks the '?' placeholder values separately.

Quality check	Result from notebook
Records	205
Original columns	15
normalized-losses missing before cleaning	41
horsepower missing before cleaning	2
price missing before cleaning	0
Missing values after median imputation	0

Cleaning approach used in the notebook

1. Replace '?' with NaN.
2. Convert normalized-losses, horsepower, and price to numeric values.
3. Fill missing numerical values with the median.
4. Verify that the dataset contains no remaining missing values.

3. Numerical & Categorical Data Preparation

The notebook separates numerical columns using pandas numeric dtype selection. The final dataset information shown in the notebook contains 205 rows and 15 columns, with 9 numerical columns and 6 categorical columns after the engineered features are included.

Skewness Analysis

The notebook reports the following skewness values. Positive skewness indicates a longer right tail in the distribution.

Feature	Skewness
symboling	0.211
normalized-losses	0.976
width	0.904
height	0.063
engine-size	1.948
horsepower	1.403
city-mpg	0.664
highway-mpg	0.540
price	1.805

Log Transformation & Feature Engineering

The notebook creates **price_log = log1p(price)** as a transformed version of the price variable. It also includes an engineered **area** feature in the final dataset. The notebook therefore extends the original 15-column structure to 15 columns after the preprocessing/feature steps shown in its final output.

4. Data Scaling & Modeling Preparation

The notebook applies **StandardScaler** to the numerical features. The final summary statistics show means close to zero and standard deviations close to one for the scaled numerical variables.

The notebook then creates X by dropping price and y as the price target. The recorded shapes are **X = (205, 14)** and **y = (205,)**.

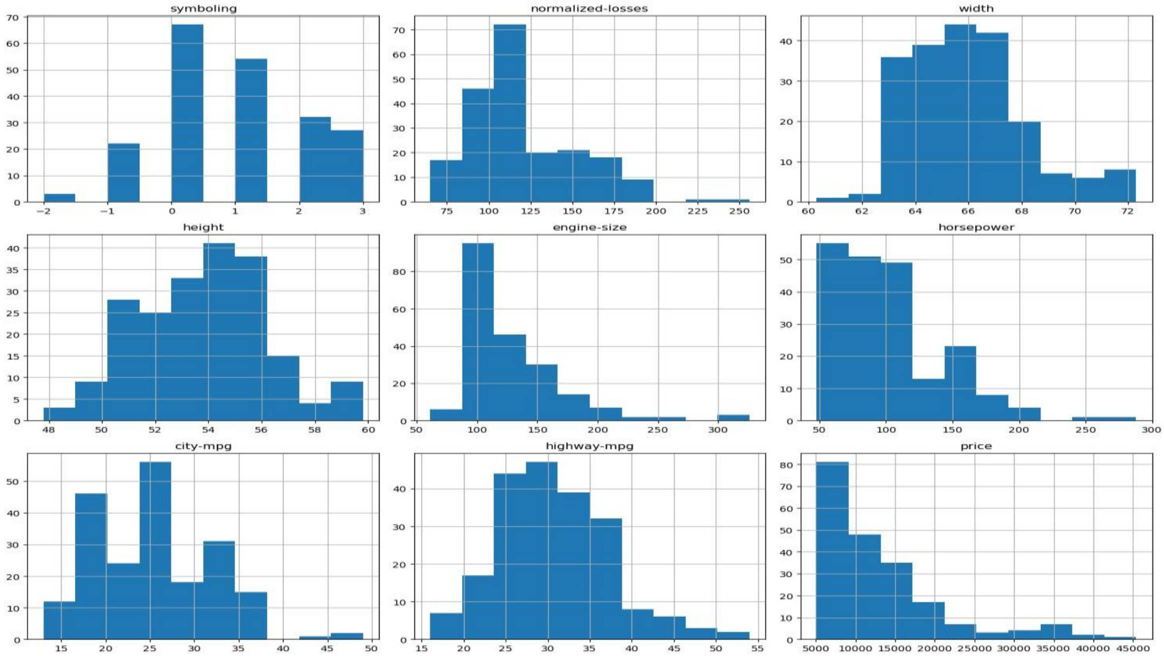
Modeling component	Notebook result
X (features)	(205, 14)
y (target)	(205,)
Target	price

Notebook interpretation note: StandardScaler is applied before the EDA visualization cells in the uploaded notebook. Therefore, the later price-based charts visualize the scaled price values rather than the original currency-scale price. This report preserves the notebook's actual outputs and does not silently replace them with recomputed charts.

5. EDA Visualizations

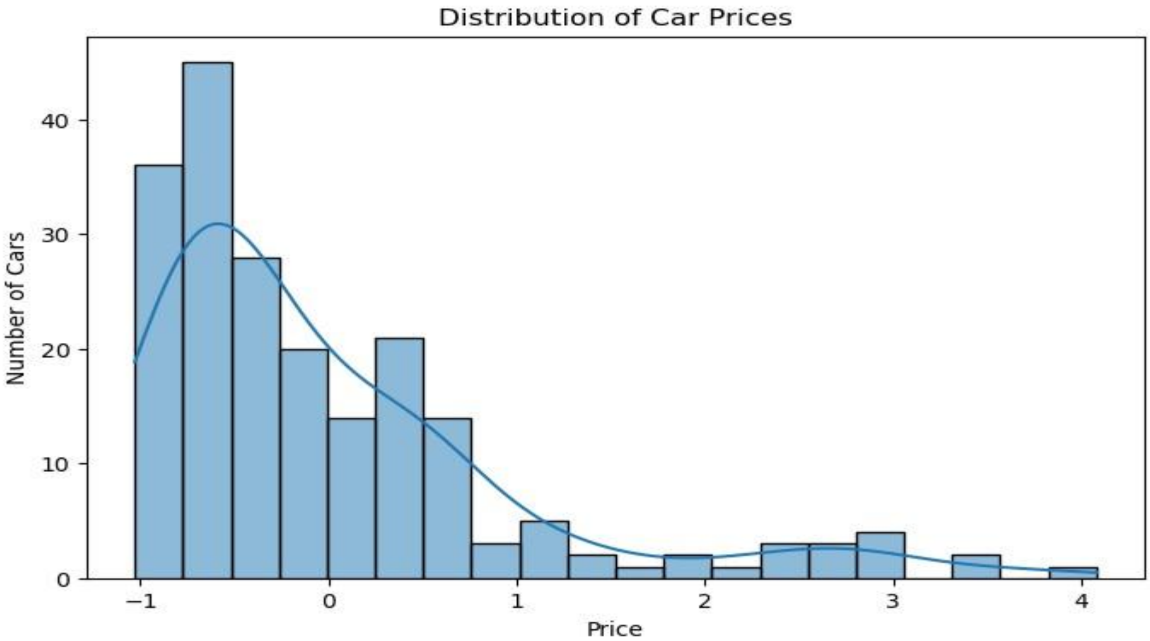
This section reproduces the charts contained in the uploaded notebook. The chart sequence follows the notebook's own Univariate, Bivariate, and Multivariate organization.

1. Histogram — Price



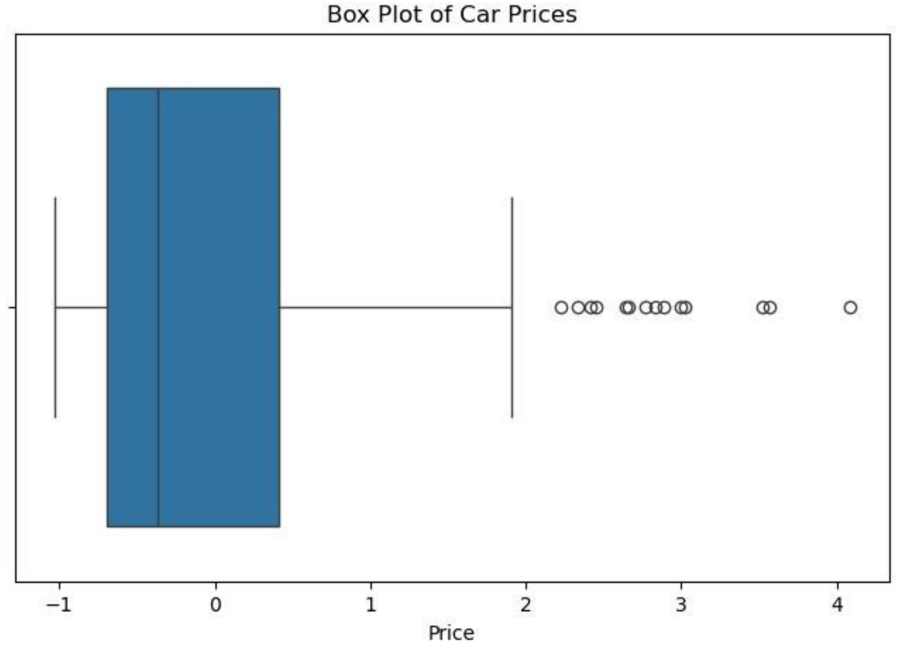
Source: uploaded project1.ipynb notebook output.

2. Box Plot — Price



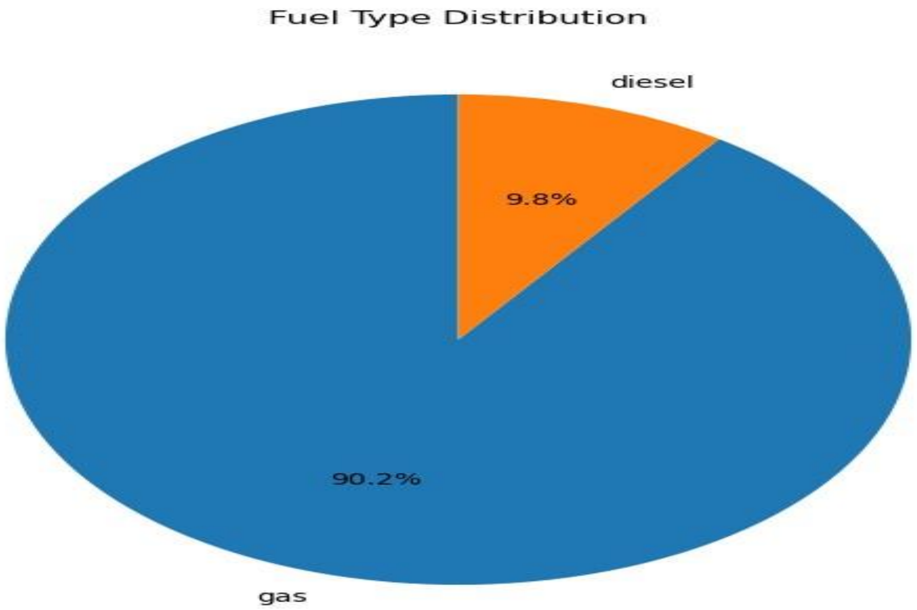
Source: uploaded project1.ipynb notebook output.

3. Pie Chart — Fuel Type



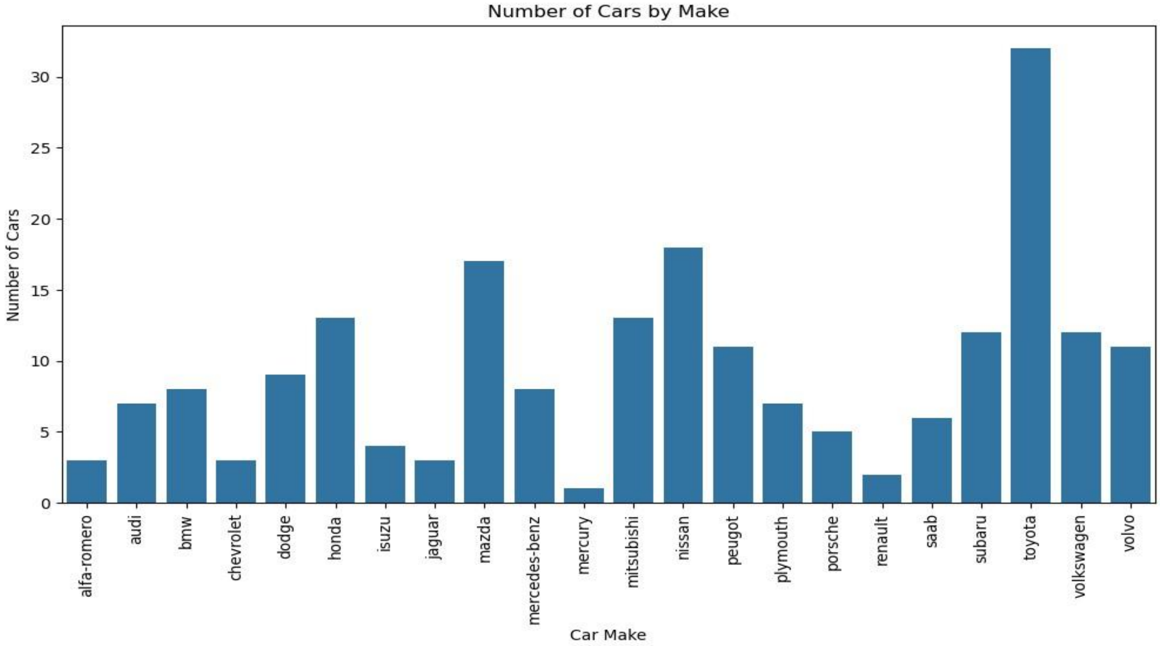
Source: uploaded project1.ipynb notebook output.

4. Count Plot — Car Make



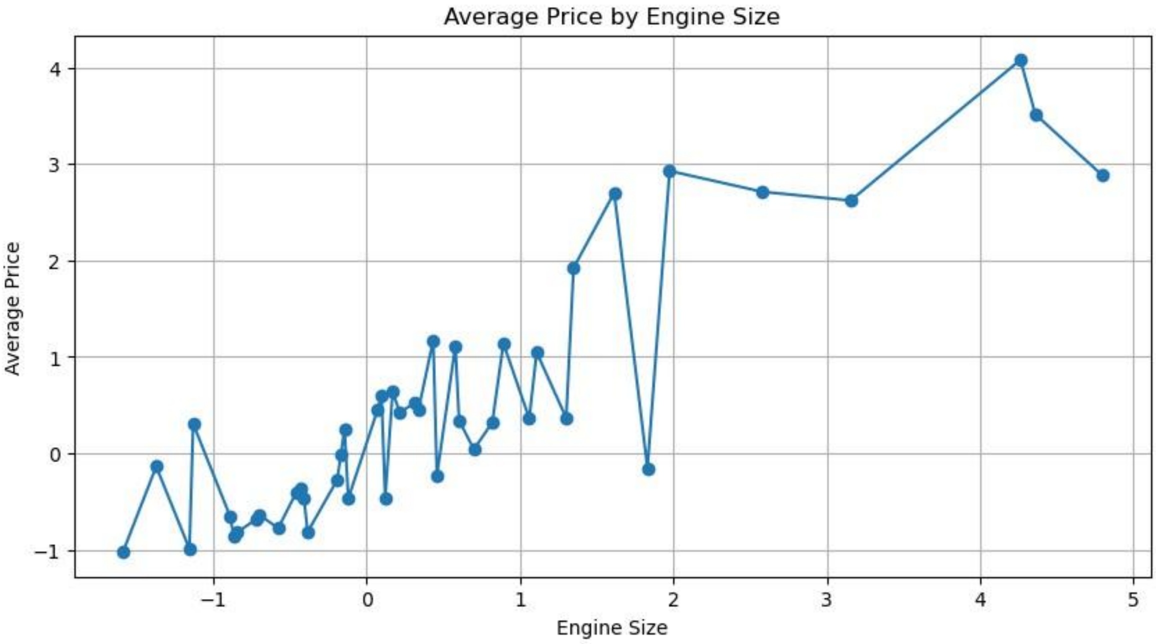
Source: uploaded project1.ipynb notebook output.

5. Line Chart — Average Price by Engine Size



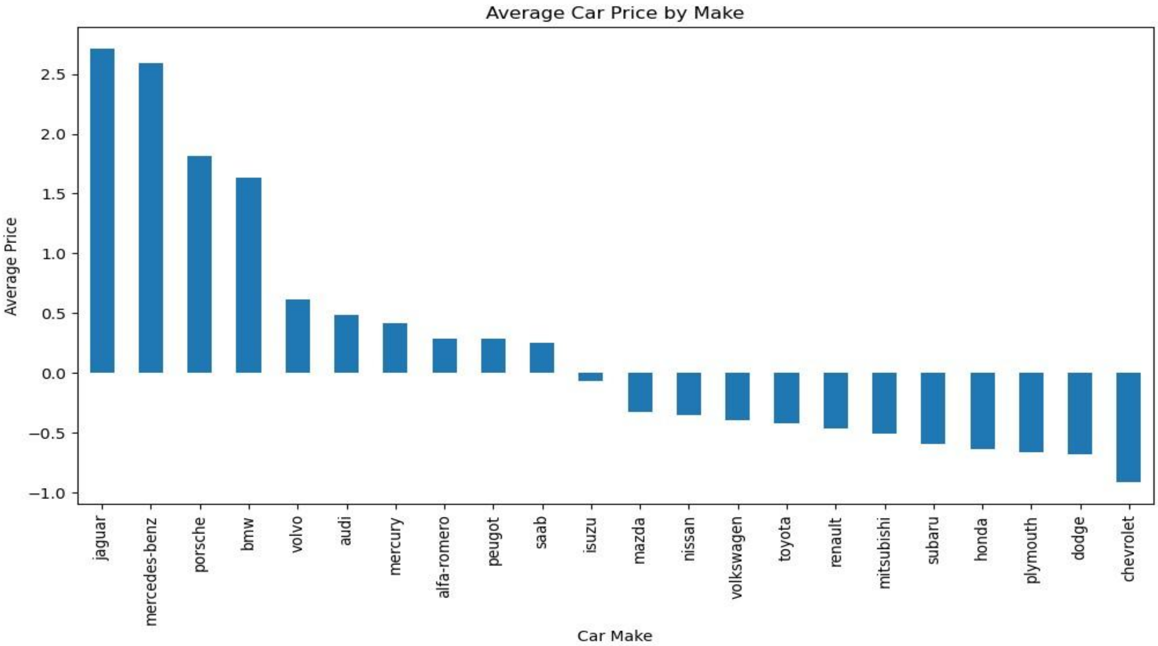
Source: uploaded project1.ipynb notebook output.

6. Bar Plot — Average Price by Make



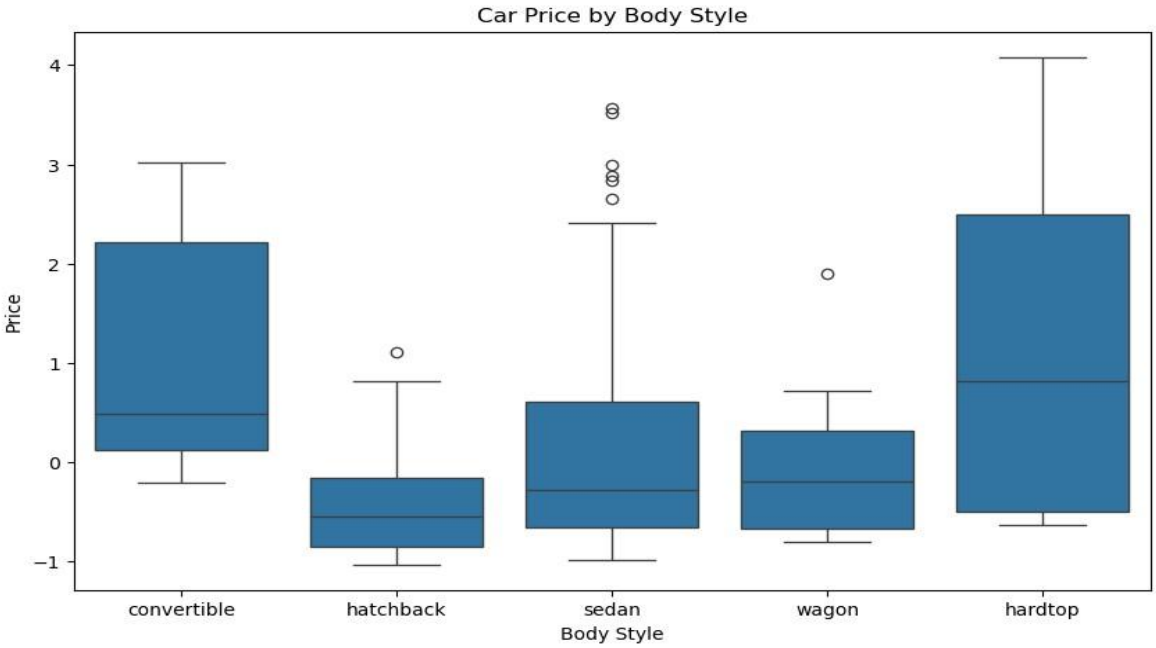
Source: uploaded project1.ipynb notebook output.

7. Box Plot — Price by Body Style



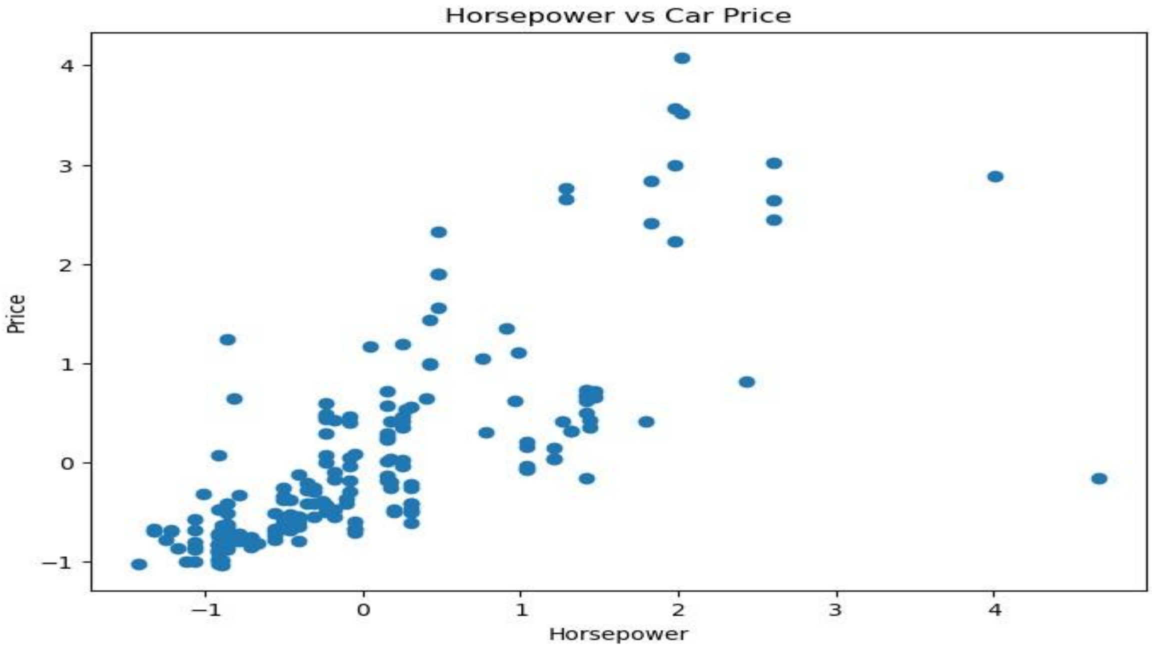
Source: uploaded project1.ipynb notebook output.

8. Scatter Plot — Horsepower vs Price



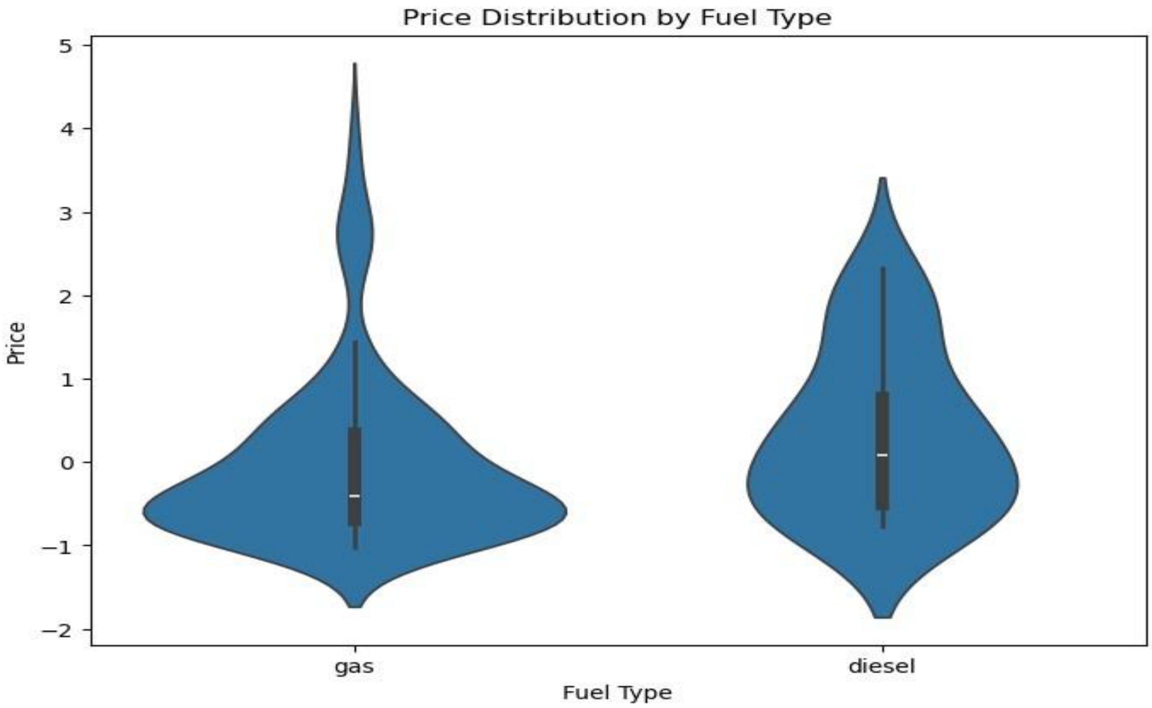
Source: uploaded project1.ipynb notebook output.

9. Violin Plot — Price by Fuel Type



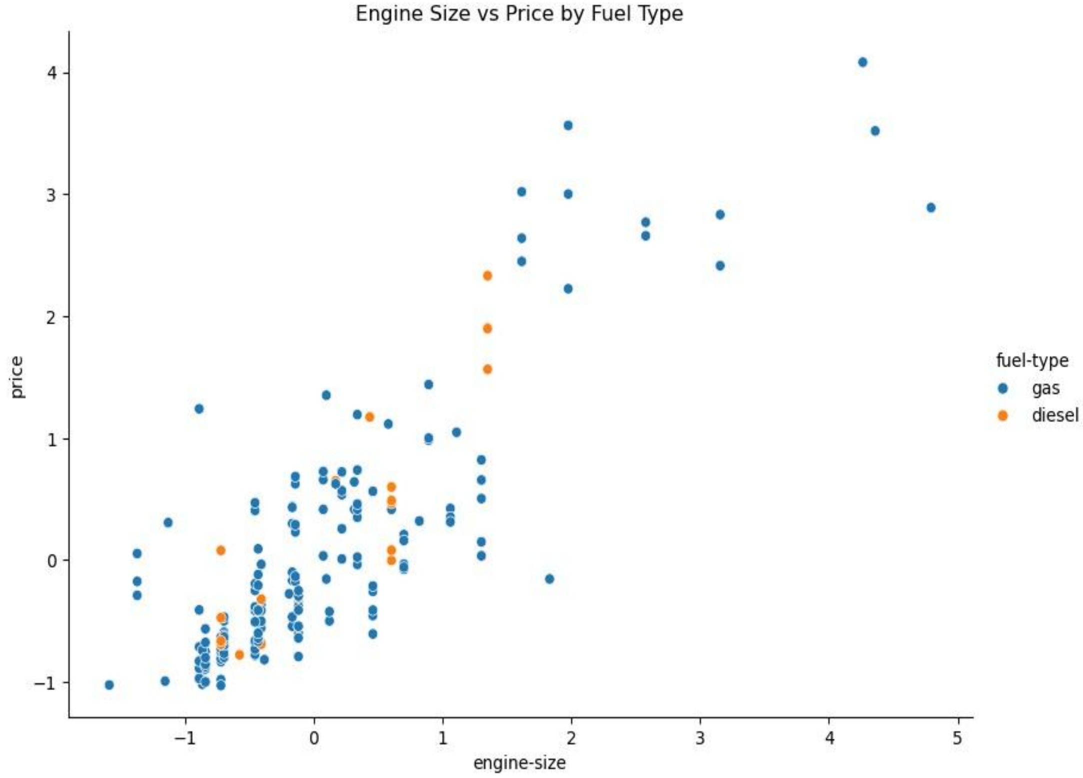
Source: uploaded project1.ipynb notebook output.

10. Relplot — Engine Size vs Price by Fuel Type



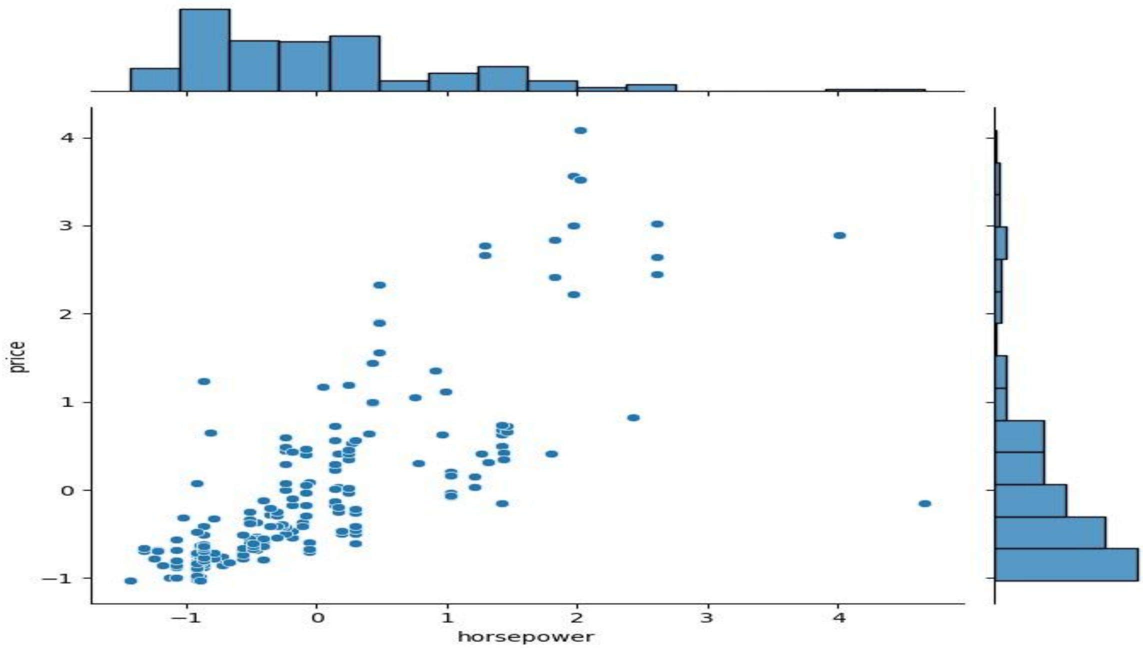
Source: uploaded project1.ipynb notebook output.

11. Joint Plot — Horsepower vs Price



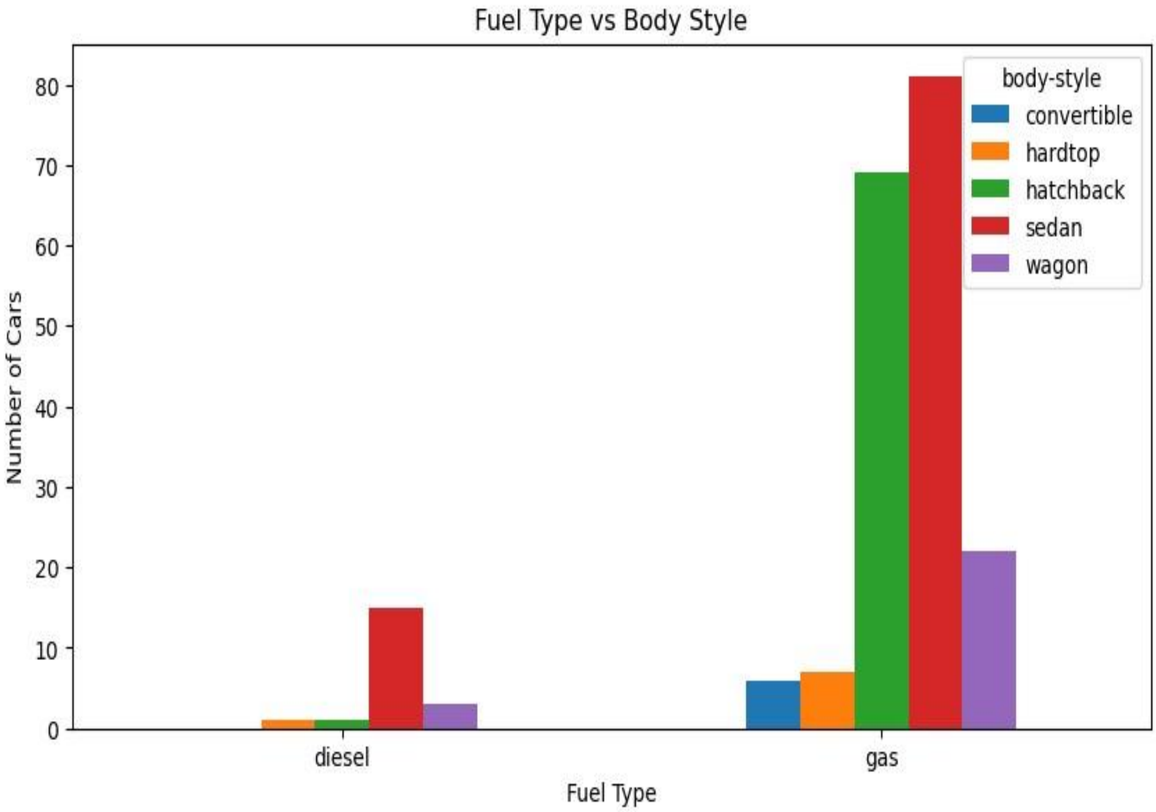
Source: uploaded project1.ipynb notebook output.

12. Crosstab — Fuel Type vs Body Style



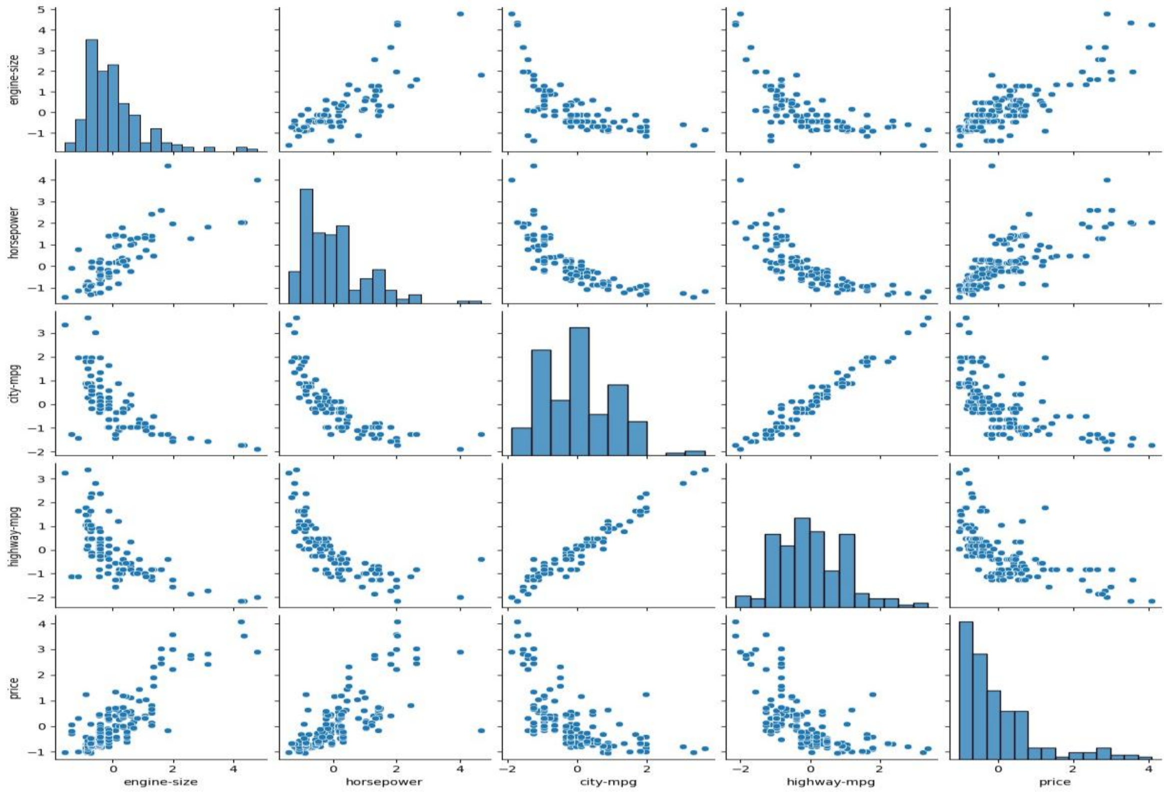
Source: uploaded project1.ipynb notebook output.

13. Pairplot



Source: uploaded project1.ipynb notebook output.

14. Correlation Heatmap



Source: uploaded project1.ipynb notebook output.

6. Bivariate Table — Fuel Type vs Body Style

The notebook produces the following crosstab between fuel type and body style:

Fuel Type	Convertible	Hardtop	Hatchback	Sedan	Wagon
Diesel	0	1	1	15	3
Gas	6	7	69	81	22

7. Key Insights & Observations

The following insights are directly derived from the analytical outputs recorded in the uploaded notebook.

- **Highest average-price makes:** Jaguar, Mercedes-Benz, Porsche, BMW, Volvo, Audi, Mercury, Alfa-Romero, Peugot, and Saab form the notebook's top-10 ranking by average scaled price.
- **Engine size:** The correlation between engine size and price is **0.853**, indicating a strong positive relationship in the notebook's scaled data.
- **Horsepower:** The correlation between horsepower and price is **0.747**, also indicating a strong positive relationship.
- **City MPG:** The correlation with price is **-0.655**, indicating an inverse relationship.
- **Highway MPG:** The correlation with price is **-0.679**, indicating an inverse relationship.
- **Drive wheels:** The notebook's average-price output ranks **RWD** highest, followed by 4WD and FWD.
- **Fuel type:** The notebook's average-price output ranks diesel above gas in the scaled values.
- **Fuel/body-style distribution:** Gas vehicles dominate the crosstab, especially sedan and hatchback categories.
- **Data quality:** 41 normalized-losses and 2 horsepower missing values were handled, leaving zero missing values in the final check.
- **Preparation:** The notebook completed scaling, feature/target separation, and visualization and states that the dataset is ready for further predictive modeling.

8. Final Summary & Conclusion

The Automobile EDA project completes a full analytical workflow beginning with dataset import and quality checking and continuing through data cleaning, numerical/categorical processing, skewness analysis, log transformation, feature engineering, scaling, feature-target separation, and exploratory visualization.

The analysis identifies meaningful relationships between price and automobile characteristics. Engine size and horsepower show positive relationships with price, while city and highway MPG show negative relationships. The project also examines manufacturer, fuel type, body style, drive wheels, and interactions between multiple numerical features.

The notebook's final status is that missing values were handled, outliers were analyzed, categorical and numerical features were processed, skewness was analyzed and transformed where required, feature engineering was performed, numerical features were scaled, and univariate, bivariate, and multivariate analysis was completed.

Project stage	Status
Dataset import & exploration	Completed
Missing-value analysis	Completed
Data cleaning	Completed
Numerical/categorical processing	Completed
Skewness analysis	Completed
Log transformation	Completed

Feature engineering	Completed
Standard scaling	Completed
Univariate analysis	Completed
Bivariate analysis	Completed
Multivariate analysis	Completed
Key insights	Completed

Conclusion: The completed EDA provides a structured understanding of the automobile dataset and prepares the feature set for further predictive modeling.